

O9820 Calibration + Find Ring Center

Purpose

This procedure validates that the probe calibration macro **O9820** can:

1. Find the ring center automatically using X-, X+, Y-, Y+ wall touches
2. Compute and store:
 - #540 = stylus effective radius
 - #541 = stylus effective length constant (Z reference)
3. Complete successfully with:
 - #585 = 1
 - #583/#584 logging the final calibration values

This is the **full "real" configuration test** after the safety and syntax tests have already passed.

Why we do these steps

Why the ring is needed

The ring provides **known geometry** (a stable inside diameter). Touching opposite walls lets the macro compute:

- The **true center** (average of opposite hit positions)
- The **effective probe radius** by comparing measured vs nominal diameter

Why we must start near the center

Center-find uses **short, bounded strokes** (software "overtravel"). If we start too far from the ring wall or too far outside the ring, the stroke may not reach the surface and the macro will alarm **921 (NO TRIGGER)**.

Why we use Safe Z and Z limits

The macros use SAFE_Z_MACHINE (#526) and Z travel limits (#524/#525) to prevent dangerous motion:

- Moves to SAFE_Z_MACHINE before XY repositioning.
- Clamps any computed Z approach plane to the configured Z limits. This prevents commanding Z to an unsafe value.

Why we need the test bar (or a real Z stack) for the Z reference portion

The Z-reference part of O9820 assumes there is a **real physical stack height** to touch:

- E = Z reference plane (machine coordinate)
- F = test bar length

The macro approaches $Z_{start} \approx E + F + \text{margin}$ and probes down to touch. If there is no real contact surface at the expected height, the Z probing will alarm 921 or produce meaningless values.

Why M19 is required before G31

This machine requires spindle orientation (M19) before skip-probing moves (G31). Without it, you may see feed/spindle state errors or probing may not execute correctly.

Safety prerequisites (must be true before running)

1. Machine is in a safe state**

- No active tool in cut
- No coolant blasting the probe (unless your process requires it)
- Door closed / interlocks satisfied
- Feed override conservative (start at 25–50% for first validation)

2. Macros initialized

- Run O9800 once after power-up to initialize defaults.
- Run O9803 once to set SAFE_Z_MACHINE.

3. Persistent variables must be set correctly

Confirm these on OFFSET → MACRO:

- #526 = SAFE_Z_MACHINE (inches, machine coordinate)
- #524 = Z_MIN limit (mm)
- #525 = Z_MAX limit (mm)
- #530 = probe seek feed (inch/min)
- #532 = probe max stroke (inch)
- #585 last status (used for pass/fail)
- (Optional logs) #580–#584

If #526 is 0, probing macros will fault (SAFE Z not set).

If #524/#525 are 0 or inverted, O9820 will fault 946 (Z LIMITS INVALID).

Required equipment

- Master ring (known inside diameter)
 - Probe in a toolholder (spindle probe)
 - Test bar tool (known length)
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Operator setup checklist

A. Mount the ring safely

1. Clean the table / fixture.
2. Place ring on vice

3. Confirm ring is square and stable (no rocking).

B. Load the probe

1. Put probe tool into spindle (manual load or ATC).
2. Ensure probe is functioning (battery/connection OK).

C. Position the probe near the ring center

1. Jog above the ring ID, roughly centered.
2. Set Z to a safe height above ring (below SAFE_Z_MACHINE is OK; you should still have clearance).

Start as close to center as possible to reduce the chance of "NO TRIGGER" alarms during wall touches.

Inputs to O9820 (what A/B/C/D/E/F mean)

You will run:

```
G65 P9820 A1 B<ring_ID> C0 D0 E<Z_ref> F<testbar_len>
```

- A1 = inch mode
 - B = ring inside diameter (inches)
 - C = ring center X machine coordinate
 - use 0 to enable automatic center-find
 - D = ring center Y machine coordinate
 - use 0 to enable automatic center-find
 - E = Z reference plane (machine Z coordinate, inches)
 - F = test bar length (inches)
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How to choose E and F (Z reference setup)

Test bar setup:

1. Identify the test bar length F (inches).
 - Use its certified value if available.
2. Decide what your Z reference plane E represents.
 - Best: E + F corresponds to a real plane the probe can touch (table surface, fixture top, or a known calibration surface).

Practical way to set E:

If you can measure a plane with the test bar and record machine Z at contact:

1. Load test bar tool.
2. Carefully touch a known plane (paper feeler / shim method).
3. Record the machine Z at that plane: call it Z_plane.
4. Set E = Z_plane - F.

This makes: • $E + F = Z_{\text{plane}}$ (a real physical plane)

Run Procedure (step-by-step)

Step 1 — First run in SINGLE BLOCK (recommended)

1. Turn on Single Block.
2. Start from a safe position above the ring.
3. Run:

```
G65 P9820 A1 B<ring_ID> C0 D0 E<Z_ref> F<testbar_len>
```

4. Watch each stage:

- Goes to G53 G0 Z#526 (safe Z)
- Approaches X-, probes X-
- Approaches X+, probes X+
- Approaches Y-, probes Y-
- Approaches Y+, probes Y+
- Returns to center
- Moves to Z start (clamped)
- Performs Z probe touch

If anything looks wrong, hit Feed Hold or E-Stop.

Step 2 — Full run (after first safe pass)

1. Turn off Single Block.
 2. Run the same command again.
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Expected results (pass criteria)

After a successful run:

- #585 = 1
- #540 updated (stylus effective radius)
- #541 updated (stylus effective length constant)
- #583 = #540 (logged)
- #584 = #541 (logged)

Sanity checks

- #540 should be reasonable:
 - Close to the stylus ball radius (or slightly different “effective” value), not huge.
- #541 should be plausible:
 - Not thousands, not near zero unexpectedly.

- Should be consistent across repeat runs if setup is repeatable.

Common alarms and what they mean

Alarm 921 — NO TRIGGER

Meaning: The probing move did not hit anything within the allowed stroke.

Causes:

- Probe not positioned close enough to ring wall before a touch.
- Stroke (#532) too small for the current setup.
- Z reference stack not present where expected.

Actions:

1. Re-position closer to the ring wall (start nearer center).
2. Confirm ring ID input B is correct.
3. Confirm probe is functional.
4. For Z portion: verify your E and F correspond to a real plane.

Alarm 946 — Z LIMITS INVALID

Meaning: Z limits are not set ($\#524/\#525 = 0$) or are inconsistent (min/max inverted).

Actions:

1. Verify #524 and #525 are set correctly (mm values).
2. Confirm conversion logic expectations (library uses #510 to interpret unit mode).

SAFE Z not set

Meaning: $\#526 = 0$

Actions:

1. Run the following command to set safe Z.

```
G65 P9803 A<safe_z_machine_in>
```

Post-run documentation (operator notes)

Record for each successful calibration run:

- Date/time
- Ring ID B used
- E and F used
- Result values:

- #540
 - #541
 - Any issues (alarms, repositioning required)
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